

Note 2: Introduction to DLA and its role

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2.1 Summary of Progress

This weekly note summarizes recent work as follows:

1. Established the fundamental framework of Dynamical Lie Algebras (DLA) and proved key structure theorems.
2. Investigated the relation between DLA and barren plateaus in parameterized quantum circuits.
3. Organized quantum circuit identities, theorems, and exercises from Nielsen & Chuang.
4. Started studying Chapters 8 and 9 of Nielsen & Chuang as scheduled.
5. Prepared for further DLA computations and the $\mathfrak{u}(N)$ decomposition proof.

2.2 Introduction to Dynamical Lie Algebra (DLA)

In quantum computation, unitary transformations (quantum gates) form the core of circuit operations. These unitary transformations collectively constitute the unitary group $U(N)$, which is a compact Lie group acting on the finite-dimensional complex Hilbert space $\mathbb{C}^N$ (with $N = 2^n$ for n qubits). A fundamental result in Lie theory is the **Lie group-Lie algebra correspondence**: every Lie group has an associated Lie algebra that describes its infinitesimal structure. For the unitary group $U(N)$, the corresponding Lie algebra is the space of skew-Hermitian matrices $\mathfrak{u}(N)$.

In practice, any quantum circuit is built from only finitely many elementary quantum gates. Each such gate corresponds to the exponential of a local Hamiltonian, which serves as an infinitesimal generator in $\mathfrak{u}(N)$. Therefore, the set of generators that span the effective dynamical structure of the circuit is also finite. Rigorous connections between gate sets, Hamiltonians, and generated Lie algebras will be established in later theorems [].

Formally, the real Lie algebra of skew-Hermitian matrices is

$$\mathfrak{u}(N) = \{A \in \mathbb{C}^{N \times N} \mid A^\dagger = -A\},$$

which serves as the infinitesimal generator space of unitary transformations in quantum systems.

Lemma 1 (Real Lie Algebra Structure of $\mathfrak{u}(N)$)

The space $\mathfrak{u}(N)$ is a real Lie algebra with commutator bracket $[A, B] = AB - BA$. A real basis is given by:

- iE_{jj} for $1 \leq j \leq N$,
- $E_{jk} - E_{kj}$ and $i(E_{jk} + E_{kj})$ for $1 \leq j < k \leq N$,

where E_{jk} are matrix units. The real dimension is

$$\dim_{\mathbb{R}} \mathfrak{u}(N) = N^2.$$

Proof The space $\mathfrak{u}(N)$ is **not a complex Lie algebra**, since multiplication by i does not preserve skew-Hermiticity.

Lie bracket closure: For any $A, B \in \mathfrak{u}(N)$,

$$[A, B]^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = BA - AB = -[A, B],$$

so $[A, B] \in \mathfrak{u}(N)$.

Span: Let $A \in \mathfrak{u}(N)$. Then $A_{jj} \in i\mathbb{R}$ and $A_{jk} = -\overline{A_{kj}}$ for $j \neq k$. Write

$$A_{jj} = i\alpha_j, \quad A_{jk} = b_{jk} + ic_{jk} \quad (j < k)$$

with real coefficients $\alpha_j, b_{jk}, c_{jk} \in \mathbb{R}$. Direct expansion shows

$$A = \sum_{j=1}^N \alpha_j iE_{jj} + \sum_{1 \leq j < k \leq N} b_{jk}(E_{jk} - E_{kj}) + \sum_{1 \leq j < k \leq N} c_{jk} i(E_{jk} + E_{kj}).$$

Thus every element in $\mathfrak{u}(N)$ lies in the real span of the proposed set.

Counting the number of generators gives

$$N + 2 \cdot \frac{N(N-1)}{2} = N^2 = \dim_{\mathbb{R}} \mathfrak{u}(N),$$

so they form a real basis of $\mathfrak{u}(N)$. ■

2.2.1 Definition and Computation Problem of DLA

Definition 2 (Dynamical Lie Algebra, DLA) Let $\mathcal{G} = \{H_1, \dots, H_k\} \subset \text{Herm}(N)$ be a finite set of Hamiltonians. The Dynamical Lie Algebra generated by $\mathcal{G}$ is defined as

$$\mathfrak{g}_{\text{DLA}} = \langle i\mathcal{G} \rangle_{\text{Lie}} \subseteq \mathfrak{u}(N).$$

Notation Convention: For any subset $S \subset \mathfrak{u}(N)$, we denote by

$$\langle S \rangle_{\text{Lie}}$$

the **minimal real Lie subalgebra** of $\mathfrak{u}(N)$ **containing** S . Since each H_j is Hermitian with zero trace, we have

$$i\mathcal{G} = \{iH_1, \dots, iH_k\} \subset \mathfrak{su}(N),$$

the Lie algebra of traceless skew-Hermitian matrices. We introduce the shorthand

$$X_j := iH_j, \quad \forall j = 1, \dots, k.$$

The problem of computing the DLA $\mathfrak{g}_{\text{DLA}}$ is therefore equivalent to finding the Lie subalgebra of $\mathfrak{su}(N)$ generated by $\{X_1, \dots, X_k\}$, i.e.,

$$\mathfrak{g}_{\text{DLA}} = \langle X_1, \dots, X_k \rangle_{\text{Lie}} \subseteq \mathfrak{su}(N).$$

Uniqueness: The minimal Lie subalgebra containing a given set is unique, so $\mathfrak{g}_{\text{DLA}}$ is uniquely determined by $\mathcal{G}$.

We formalize three distinct levels of computing the DLA as follows.

Definition 3 (DLA Dimension Computation) Given $\mathcal{G} = \{H_1, \dots, H_k\} \subset \text{Herm}(N)$ and its DLA $\mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}$, the **DLA dimension computation** asks only for the real dimension

$$\dim_{\mathbb{R}} \mathfrak{g}.$$

Definition 4 (DLA Basis / Closure Computation) Given $\mathcal{G} = \{H_1, \dots, H_k\} \subset \text{Herm}(N)$ and its DLA $\mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}$, the *DLA basis computation* (closure computation) finds a real basis $\{Y_1, \dots, Y_d\} \subset \mathfrak{u}(N)$ such that

$$\mathfrak{g} = \text{span}_{\mathbb{R}} \{Y_1, \dots, Y_d\}.$$

This is the standard form of computing the DLA in the literature.

Definition 5 (DLA Structure Identification) Given $\mathcal{G} = \{H_1, \dots, H_k\} \subset \text{Herm}(N)$ and its DLA $\mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}$, **DLA structure identification** determines the Lie-algebra isomorphism class of $\mathfrak{g}$, including its decomposition into center and simple summands:

$$\mathfrak{g} \cong \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{s}_1 \oplus \dots \oplus \mathfrak{s}_m.$$

We further distinguish two computational paradigms:

- *Symbolic DLA computation:* Uses exact commutation relations and symbolic matrix expressions, yielding error-free results.
- *Numerical DLA computation:* Uses floating-point matrix representations and numerical rank detection, trading precision for speed.

2.2.2 Construction Theorem for DLA

The following result gives an explicit inductive construction of the DLA by iteratively closing the generating set under the Lie bracket.

Theorem 6 ((DLA Construction via Inductive Closure):) Let $\{X_1, \dots, X_k\} \subset \mathfrak{u}(N)$ be a finite set of skew-Hermitian generators. Define an inductive sequence of real subspaces of $\mathfrak{u}(N)$:

$$\begin{cases} \mathfrak{g}_0 = \text{span}_{\mathbb{R}} \{X_1, X_2, \dots, X_k\} \\ \mathfrak{g}_{m+1} = \mathfrak{g}_m + \text{span}_{\mathbb{R}} \{[A, B] \mid A, B \in \mathfrak{g}_m\} \quad (\forall m \geq 0) \end{cases}$$

where $\text{span}_{\mathbb{R}} S$ denotes the real linear span of a set S . Then:

1. The sequence stabilizes: There exists a minimal integer $M \geq 0$ such that $\mathfrak{g}_{M+1} = \mathfrak{g}_M$;
2. The stable subspace $\mathfrak{g}_M$ is exactly the DLA $\mathfrak{g}_{\text{DLA}}$ (satisfies all four axioms of the axiomatic definition).

Proof

1. **Stabilization:** The sequence $\mathfrak{g}_0 \subseteq \mathfrak{g}_1 \subseteq \dots$ is an increasing chain of real subspaces in $\mathfrak{u}(N)$. Since $\dim_{\mathbb{R}} \mathfrak{u}(N) = N^2 < \infty$, the chain must terminate after finitely many steps. Thus there exists a minimal $M \geq 0$ such that $\mathfrak{g}_{M+1} = \mathfrak{g}_M$.

2. **Identification with the DLA:** Let $\mathfrak{g} = \mathfrak{g}_M$. We verify that $\mathfrak{g}$ is the dynamical Lie algebra:

- **Inheritance of generators:** By construction, $\{X_1, \dots, X_k\} \subseteq \mathfrak{g}_0 \subseteq \mathfrak{g}$.
- **Closure under Lie bracket:** Since $\mathfrak{g}_{M+1} = \mathfrak{g}_M$, we have

$$[\mathfrak{g}, \mathfrak{g}] = \text{span}_{\mathbb{R}} \{[A, B] : A, B \in \mathfrak{g}\} \subseteq \mathfrak{g}.$$

Hence $\mathfrak{g}$ is a real Lie subalgebra of $\mathfrak{u}(N)$.

- **Minimality:** Let $\mathfrak{h} \subseteq \mathfrak{u}(N)$ be any real Lie subalgebra containing $\{X_1, \dots, X_k\}$. By induction, $\mathfrak{g}_m \subseteq \mathfrak{h}$ for all m , so $\mathfrak{g} \subseteq \mathfrak{h}$.

Therefore $\mathfrak{g}$ is the unique minimal real Lie subalgebra containing the generators, i.e., $\mathfrak{g} = \mathfrak{g}_{\text{DLA}}$. ■

2.2.3 Unitary Groups Generated by DLA

This fundamental correspondence connects the Dynamical Lie Algebra (a Lie subalgebra of infinitesimal generators) to the connected unitary group realized by quantum circuits composed of the corresponding elementary gates. It justifies why computing the DLA completely characterizes the dynamical capabilities of a given quantum gate set.

Lemma 7 (Commutator Exponential Formula) *For any $A, B \in \mathfrak{u}(N)$ and $t \in \mathbb{R}$, the exponential of the commutator can be approximated by products of exponentials:*

$$\exp(t[A, B]) = \lim_{n \rightarrow \infty} \left(\exp\left(\frac{t}{n}A\right) \exp\left(\frac{t}{n}B\right) \exp\left(-\frac{t}{n}A\right) \exp\left(-\frac{t}{n}B\right) \right)^{n^2},$$

where $\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$ denotes the matrix exponential.

Theorem 8 *Let $\mathcal{S} \subset \mathfrak{u}(N)$ be a finite subset, then we have*

$$\exp(\langle \mathcal{S} \rangle_{\text{Lie}}) = \overline{\langle \exp X : X \in \mathcal{S} \rangle}.$$

as Lie subgroup of $U(N)$.

Proof Let

$$\mathfrak{g} := \langle \mathcal{S} \rangle_{\text{Lie}}, \quad G_1 := \exp(\mathfrak{g}), \quad G_2 := \langle \exp X : X \in \mathcal{S} \rangle.$$

We prove the double inclusion $G_1 = \overline{G_2}$.

1. $\overline{G_2} \subseteq G_1$ By construction, $\mathfrak{g}$ is a real Lie subalgebra of $\mathfrak{u}(N)$, so $G_1 = \exp(\mathfrak{g})$ is a connected Lie subgroup of $U(N)$. Since $X \in \mathcal{S} \subset \mathfrak{g}$, we have $\exp(X) \in G_1$ for all $X \in \mathcal{S}$. Thus $G_2 \subseteq G_1$. As G_1 is closed in $U(N)$ (being a compact Lie subgroup), it follows that $\overline{G_2} \subseteq G_1$.

2. $G_1 \subseteq \overline{G_2}$ By the commutator exponential lemma, for any $A, B \in \mathfrak{g}$ and $t \in \mathbb{R}$,

$$\exp(t[A, B]) \in \overline{G_2}.$$

Since $\mathfrak{g}$ is generated by $\mathcal{S}$ under Lie brackets, every element of $\exp(\mathfrak{g})$ can be approximated by products of exponentials of elements in $\mathcal{S}$ and their iterated commutators. Hence $G_1 = \exp(\mathfrak{g}) \subseteq \overline{G_2}$.

Combining both inclusions yields $\exp(\langle \mathcal{S} \rangle_{\text{Lie}}) = \overline{\langle \exp X : X \in \mathcal{S} \rangle}$. ■

2.2.4 The Complexification of $\mathfrak{u}(N)$

$\mathfrak{u}(N)$ (the Lie algebra of $N \times N$ skew-Hermitian matrices) is a compact real Lie algebra, and its structure is characterized by two core properties: reducibility (direct sum of center and semisimple part) and complexification isomorphism to the full complex matrix algebra.

Theorem 9 (Complexification Isomorphism of $\mathfrak{u}(N)$): *There exists an isomorphism of complex Lie algebras:*

$$\mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{gl}(N, \mathbb{C}),$$

where $\mathfrak{gl}(N, \mathbb{C}) = \mathbb{C}^{N \times N}$ (full complex matrix algebra) with Lie bracket $[M, N] = MN - NM$.

Proof Define the map

$$\begin{aligned} \varphi : \mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C} &\rightarrow \mathfrak{gl}(N, \mathbb{C}) \\ \varphi \left(\sum_{j=1}^t A_j \otimes z_j \right) &= \sum_{j=1}^t z_j A_j \end{aligned}$$

1. Well-definedness φ is compatible with the real tensor product relations:

Additivity:

$$\varphi((A_1 \otimes z_1) + (A_2 \otimes z_2)) = z_1 A_1 + z_2 A_2 = \varphi(A_1 \otimes z_1) + \varphi(A_2 \otimes z_2),$$

Real scalar compatibility: for all $r \in \mathbb{R}$

$$\varphi((rA) \otimes z) = z(rA) = r(zA) = \varphi(A \otimes (rz)).$$

2. $\mathbb{C}$ -linearity For any $w, z \in \mathbb{C}$ and $A \in \mathfrak{u}(N)$:

$$\varphi(w \cdot (A \otimes z)) = \varphi(A \otimes (wz)) = wzA = w(zA) = w\varphi(A \otimes z).$$

3. Surjectivity Any $M \in \mathfrak{gl}(N, \mathbb{C})$ decomposes into skew-Hermitian and Hermitian parts:

$$M = \underbrace{\frac{M - M^\dagger}{2}}_A + i \cdot \underbrace{\frac{M + M^\dagger}{2i}}_B,$$

where $A^\dagger = -A$ and $B^\dagger = B$ (hence $A, B \in \mathfrak{u}(N)$). Thus $M = \varphi(A \otimes 1 + B \otimes i)$, so φ is surjective.

4. Injectivity The complex dimension of both spaces equals N^2 :

$$\dim_{\mathbb{C}}(\mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C}) = \dim_{\mathbb{R}} \mathfrak{u}(N) = N^2 = \dim_{\mathbb{C}} \mathfrak{gl}(N, \mathbb{C}).$$

Surjective linear maps between finite-dimensional complex vector spaces of equal dimension are injective.

5. Lie Bracket Preservation For pure tensors $A \otimes z, B \otimes w \in \mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C}$:

$$\varphi([A \otimes z, B \otimes w]) = \varphi([A, B] \otimes zw) = zw[A, B] = [zA, wB] = [\varphi(A \otimes z), \varphi(B \otimes w)].$$

By $\mathbb{C}$ -linearity, this extends to all elements of the tensor product.

Thus φ is a complex Lie algebra isomorphism. ■

Theorem 10 (Structure of Dynamical Lie Algebras) *Every dynamical Lie algebra $\mathfrak{g} = \mathfrak{g}_{\text{DLA}} \subseteq \mathfrak{u}(N)$ admits a direct sum decomposition*

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \cdots \oplus \mathfrak{g}_{k-1} \oplus \mathfrak{z}(\mathfrak{g}),$$

where $\mathfrak{g}_1, \dots, \mathfrak{g}_{k-1}$ are simple Lie algebra components and $\mathfrak{z}(\mathfrak{g})$ is the abelian center of $\mathfrak{g}$.

Remark (Relevance to DLA): Since the DLA $\mathfrak{g}_{\text{DLA}} = \langle X_1, \dots, X_k \rangle_{\text{Lie}} \subseteq \mathfrak{u}(N)$ (from our notation convention), its structure is inherited from $\mathfrak{u}(N)$: as a subalgebra of a simple Lie algebra, $\mathfrak{g}_{\text{DLA}}$ is itself semisimple (or trivial), and its complexification $\mathfrak{g}_{\text{DLA}} \otimes_{\mathbb{R}} \mathbb{C}$ is a subalgebra of $\mathfrak{gl}(N, \mathbb{C})$. This simplifies the structural identification of DLA via **complex semisimple Lie algebra classification**.

2.3 Parameterized Quantum Circuits and BP

Let $n \in \mathbb{N}$ be the number of qubits, and let $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ be the associated Hilbert space with dimension $N = 2^n$. Denote by $\mathfrak{u}(N)$ the Lie algebra of $N \times N$ skew-Hermitian matrices, and by $\mathrm{U}(N)$ the unitary group on $\mathcal{H}$.

2.3.1 Mathematical Formalism

Definition 11 A *parameterized quantum circuit (PQC)* is a unitary operator

$$U(\theta): \mathbb{R}^L \rightarrow \mathrm{U}(N), \quad \theta \mapsto U(\theta) = \prod_{\ell=1}^L e^{iH_\ell \theta_\ell},$$

where $L \in \mathbb{N}$ is the circuit depth, $\theta = (\theta_1, \dots, \theta_L) \in \mathbb{R}^L$ is the parameter vector, and $H_\ell \in i\mathfrak{u}(N)$ are Hermitian generators. The set of generators is denoted $\mathcal{G} = \{H_1, \dots, H_L\}$.

2.3.2 Variational Loss Function and BP

Let ρ be a density matrix on $\mathcal{H}$, and let $O \in i\mathfrak{u}(N)$ be a Hermitian observable with $\|O\|_2^2 \leq N$.

Definition 12 The variational loss function of the PQC is

$$\ell_\theta(\rho, O) = \mathrm{Tr}(U(\theta)\rho U(\theta)^\dagger O).$$

The variance of ℓ_θ over the parameter space is

$$\mathrm{Var}_\theta[\ell_\theta(\rho, O)] = \mathbb{E}_\theta[\ell_\theta(\rho, O)^2] - (\mathbb{E}_\theta[\ell_\theta(\rho, O)])^2.$$

Definition 13 (Barren Plateau (BP)) We say the loss function exhibits a **barren plateau** if its variance vanishes exponentially with the number of qubits n , i.e.,

$$\mathrm{Var}_\theta[\ell_\theta(\rho, O)] \in \mathcal{O}(1/b^n)$$

for some constant $b > 1$.

Definition Origin: g-Purity

First formally defined in: *A Lie algebraic theory of barren plateaus for deep parameterized quantum circuits* M. Ragone, B. Bakalov, A. Fontalvo, J. I. Costa, A. Cerezo (2023)
arXiv:2309.09342, Nature Communications (2024)

Definition 14 g-Purity (Generalized Purity)

- Let $\mathfrak{g} \subseteq \mathfrak{u}(2^n)$ be a real operator subalgebra, with complexification $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$.
- Let $\{B_j\}_{j=1}^{\dim(\mathfrak{g})}$ be an orthonormal basis of $\mathfrak{g}_{\mathbb{C}}$ under the inner product $\langle A, B \rangle = \mathrm{Tr}[A^\dagger B]$.

For any Hermitian operator $H \in i\mathfrak{u}(2^n)$, the **g-purity** of H is defined as

$$\mathcal{P}_{\mathfrak{g}}(H) = \mathrm{Tr}[H_{\mathfrak{g}}^2] = \sum_{j=1}^{\dim(\mathfrak{g})} \left| \mathrm{Tr}[B_j^\dagger H] \right|^2,$$

where $H_{\mathfrak{g}}$ denotes the orthogonal projection of H onto $\mathfrak{g}_{\mathbb{C}}$.

$\mathcal{P}_{\mathfrak{g}}(H)$ quantifies the fraction of H contained in the subalgebra $\mathfrak{g}$:

- $\mathcal{P}_{\mathfrak{g}}(H) = \text{Tr}[H^2]$ if and only if $H \in i\mathfrak{g}$ (full alignment with $\mathfrak{g}$);
- $\mathcal{P}_{\mathfrak{g}}(H) = 0$ if and only if H is orthogonal to $\mathfrak{g}$ (no overlap with $\mathfrak{g}$);
- The standard purity $\text{Tr}[H^2]$ is recovered by taking $\mathfrak{g} = \mathfrak{u}(2^n)$.

Theorem 15 (McClean et al., 2018) *For random PQCs forming a 2-design, the loss variance decays exponentially:*

$$\text{Var}_{\theta} [\ell_{\theta}(\rho, O)] \in O(c^{-n}), \quad c > 1.$$

This is the first rigorous barren plateau result.

Theorem 16 (Hayes et al., 2022) *The loss variance is bounded by the DLA dimension:*

$$\text{Var}_{\theta} [\ell_{\theta}(\rho, O)] \lesssim \frac{\text{Tr}[\rho^2] \text{Tr}[O^2]}{\dim \mathfrak{g}_{\text{DLA}}}.$$

First connection between DLA size and barren plateaus.

Theorem 17 (Ragone et al., 2023 (early)) *Under Haar averaging, the variance scales inversely with total DLA dimension:*

$$\text{Var}_{\theta} [\ell_{\theta}(\rho, O)] = \frac{\alpha}{\dim \mathfrak{g}_{\text{DLA}}},$$

where α is an alignment constant (no purity, no decomposition).

This paper: M. Ragone et al. (2023), arXiv:2309.09342

Theorem 18 (Ragone et al., 2023) *Let $\mathfrak{g} = \mathfrak{g}_{\text{DLA}}$ be the dynamical Lie algebra with decomposition*

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_{k-1} \oplus \mathfrak{g}_k,$$

where $\mathfrak{g}_1, \dots, \mathfrak{g}_{k-1}$ are simple components and $\mathfrak{g}_k$ is the abelian center.

Suppose $O \in i\mathfrak{g}$ or $\rho \in i\mathfrak{g}$. Then

1. *The expectation of $\ell_{\theta}(\rho, O)$ is*

$$\mathbb{E}_{\theta} [\ell_{\theta}(\rho, O)] = \text{Tr} [\rho_{\mathfrak{g}_k} O_{\mathfrak{g}_k}].$$

2. *The variance of $\ell_{\theta}(\rho, O)$ is*

$$\text{Var}_{\theta} [\ell_{\theta}(\rho, O)] = \sum_{j=1}^{k-1} \frac{\mathcal{P}_{\mathfrak{g}_j}(\rho) \mathcal{P}_{\mathfrak{g}_j}(O)}{\dim(\mathfrak{g}_j)}.$$

This is the first exact formula with purity and Levi decomposition.

Corollary Origin: Barren Plateau Sufficient Condition

Corollary 19 (Ragone et al., 2023) *Let $\|O\|_2^2 \leq 2^n$.*

If either $\dim(\mathfrak{g})$, $1/\mathcal{P}_{\mathfrak{g}}(\rho)$ or $1/\mathcal{P}_{\mathfrak{g}}(O)$ is in $\Omega(b^n)$ with $b > 2$, then the loss exhibits a barren plateau (BP).

Table 2.1: Three sources of barren plateaus (Lie algebraic framework)

BP Source	Condition	Implication
(i) Circuit expressiveness (DLA)	$\dim(\mathfrak{g}) = \Omega(b^n)$, $b > 2$	Exponential DLA $\rightarrow$ severe BP, untrainable
(ii) Initial state ρ	$\mathcal{P}_{\mathfrak{g}}(\rho) = O(1/b^n)$	Low $\mathfrak{g}$ -purity $\rightarrow$ BP even with polynomial DLA
(iii) Measurement operator O	$\mathcal{P}_{\mathfrak{g}}(O) = O(1/b^n)$	Poor alignment $\rightarrow$ BP even with good circuit/state

Table 2.2: Trade-off between DLA dimension, expressibility and trainability

Property	Small DLA dimension	Exponential large DLA dimension
Expressibility	Low, insufficient for target tasks	High, strong representation capacity
Variance of loss $\text{Var} \sim 1/\dim \mathfrak{g}$	Large, stable optimization	Exponentially small, vanishing gradient
Barren plateau	Absent, good trainability	Severe, optimization fails
Practical implication	Easy to train but weak model	Powerful model but untrainable

Target: Polynomial DLA Dimension

We aim for

$$\dim \mathfrak{g} = O(n^k)$$

where n is the number of qubits and k is a small constant. This ensures balanced expressibility and trainability, avoiding barren plateaus.

2.4 Plan for Next Week

Focus on the subsequent chapters of Nielsen & Chuang:

1. **Chapter Studies:** Study standard advanced content, organize frameworks and supplement detailed proofs for textbook derivations.
 - Chapters 8, 9: **by April 17**
 - Chapters 3, 5: **by April 24**
 - First half of Chapter 10: **by May 1**
 - Second half of Chapter 10: **by May 8**
 - Chapter 6: **by May 15**
2. **DLA Research:** Study theoretical and dimensional methods for Dynamical Lie Algebras, and perform computations or numerical simulations for typical examples.
3. **Supplementary Work:** Complete exercises in the appendix and prove the decomposition theorem for the unitary Lie algebra $\mathfrak{u}(N)$.

2.5 Appendix

2.5.1 Some identities and theorem of quantum circuits

The rotation operators are defined by

$$\begin{aligned} R_x(\theta) &\equiv e^{-i\theta X/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} X = \begin{bmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}, \\ R_y(\theta) &\equiv e^{-i\theta Y/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Y = \begin{bmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix}, \\ R_z(\theta) &\equiv e^{-i\theta Z/2} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Z = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}. \end{aligned}$$

Other identities

$$H R_z(\theta) H = R_x(\theta), \quad S R_x(\theta) S^\dagger = R_z(\theta).$$

Theorem 20 For two unitary operator , we define

$$E(U, V) := \max_{|\psi\rangle} \| (U - V)|\psi\rangle \|$$

then

$$E(U_m U_{m-1} \cdots U_1, V_m V_{m-1} \cdots V_1) \leq \sum_{j=1}^m E(U_j, V_j).$$

Theorem 21 (*Z-Y decomposition for a single qubit*)

Suppose U is a unitary operation on a single qubit. Then there exist real numbers α, β, γ and δ such that

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta).$$

Proof Since U is unitary, the rows and columns of U are orthonormal, from which it follows that there exist real numbers α, β, γ , and δ such that

$$U = \begin{bmatrix} e^{i(\alpha-\beta/2-\delta/2)} \cos \frac{\gamma}{2} & -e^{i(\alpha-\beta/2+\delta/2)} \sin \frac{\gamma}{2} \\ e^{i(\alpha+\beta/2-\delta/2)} \sin \frac{\gamma}{2} & e^{i(\alpha+\beta/2+\delta/2)} \cos \frac{\gamma}{2} \end{bmatrix}.$$

■

Theorem 22 Suppose $\hat{m}$ and $\hat{n}$ are non-parallel real unit vectors in three dimensions. Use Theorem 4.1 to show that an arbitrary single qubit unitary U may be written

$$U = e^{i\alpha} R_{\hat{n}}(\beta) R_{\hat{m}}(\gamma) R_{\hat{n}}(\delta),$$

for appropriate choices of α, β, γ and δ .

Theorem 23 Suppose U is a unitary gate on a single qubit. Then there exist unitary operators A, B, C

on a single qubit such that

$$ABC = I, \quad U = e^{i\alpha}AXBXC,$$

where α is some overall phase factor.

Proof In the notation of Theorem 4.1, set $A \equiv R_z(\beta)R_y(\gamma/2)$, $B \equiv R_y(-\gamma/2)R_z(-(\delta + \beta)/2)$ and $C \equiv R_z((\delta - \beta)/2)$. Note that

$$ABC = R_z(\beta)R_y\left(\frac{\gamma}{2}\right)R_y\left(-\frac{\gamma}{2}\right)R_z\left(-\frac{\delta + \beta}{2}\right)R_z\left(\frac{\delta - \beta}{2}\right) = I. \quad (4.14)$$

Since $X^2 = I$, and using Exercise 4.7, we see that

$$XBX = XR_y\left(-\frac{\gamma}{2}\right)XXR_z\left(-\frac{\delta + \beta}{2}\right)X = R_y\left(\frac{\gamma}{2}\right)R_z\left(\frac{\delta + \beta}{2}\right). \quad (4.15)$$

Thus

$$AXBXC = R_z(\beta)R_y\left(\frac{\gamma}{2}\right)R_y\left(\frac{\gamma}{2}\right)R_z\left(\frac{\delta + \beta}{2}\right)R_z\left(\frac{\delta - \beta}{2}\right) \quad (4.16)$$

$$= R_z(\beta)R_y(\gamma)R_z(\delta). \quad (4.17)$$

Thus $U = e^{i\alpha}AXBXC$ and $ABC = I$, as required. $\square$

2.5.2 Some exercises and problems

Exercise 4.12: Give A, B, C , and α for the Hadamard gate.

Problem 2.1: (Functions of the Pauli matrices) Let $f \in H(\mathbb{C})$ be an analytic function, and $\vec{n}$ be a normalized vector in three dimensions, and let θ be real. Show that

$$f(\theta\vec{n} \cdot \vec{\sigma}) = \frac{f(\theta) + f(-\theta)}{2}I + \frac{f(\theta) - f(-\theta)}{2}\vec{n} \cdot \vec{\sigma}.$$

Problem 2.2: (Properties of the Schmidt number) Suppose $|\psi\rangle$ is a pure state of a composite system with components A and B .

(1) Prove that the Schmidt number of $|\psi\rangle$ is equal to the rank of the reduced density matrix $\rho_A \equiv \text{tr}_B(|\psi\rangle\langle\psi|)$. (Note that the rank of a Hermitian operator is equal to the dimension of its support.)

(2) Suppose $|\psi\rangle = \sum_j |\alpha_j\rangle|\beta_j\rangle$ is a representation for $|\psi\rangle$, where $|\alpha_j\rangle$ and $|\beta_j\rangle$ are (un-normalized) states for systems A and B , respectively. Prove that the number of terms in such a decomposition is greater than or equal to the Schmidt number of $|\psi\rangle$, $\text{Sch}(\psi)$.

(3) Suppose $|\psi\rangle = \alpha|\varphi\rangle + \beta|\gamma\rangle$. Prove that

$$\text{Sch}(\psi) \geq |\text{Sch}(\varphi) - \text{Sch}(\gamma)|.$$

Problem 2.3: (Tsirelson's inequality) Suppose $Q = \vec{q} \cdot \vec{\sigma}$, $R = \vec{r} \cdot \vec{\sigma}$, $S = \vec{s} \cdot \vec{\sigma}$, $T = \vec{t} \cdot \vec{\sigma}$, where $\vec{q}, \vec{r}, \vec{s}$ and $\vec{t}$ are real unit vectors in three dimensions. Show that

$$(Q \otimes S + R \otimes S + R \otimes T - Q \otimes T)^2 = 4I + [Q, R] \otimes [S, T].$$

Use this result to prove that

$$\langle Q \otimes S \rangle + \langle R \otimes S \rangle + \langle R \otimes T \rangle - \langle Q \otimes T \rangle \leq 2\sqrt{2}.$$

References

- [1] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*, 10th Anniversary Edition. Cambridge University Press, 2010.